



KiCad Design Rules & Constraints

DRC custom rules • Net classes • Layer-specific rules • Impedance constraints

Codify your design intent so KiCad enforces it

Audience: Intermediate KiCad Users

Net classes • Custom DRC • Constraints • Coupling • Layer rules



Why Codify Rules

Catch errors early

DRC enforces rules
at design time, not fab.

Consistency

Same rules applied
across the project.

Team alignment

Rules baked into project
— not in someone's head.

Fab capability

Match fab's min trace +
drill + clearance.

Performance

High-speed nets need
tighter constraints.

Documentation

Rules file = explicit
specification.

DRC Categories

- ▶ Clearance violations: copper too close to other copper
- ▶ Trace width: too thin (below min spec)
- ▶ Drill hole: too small (below fab capability)
- ▶ Annular ring: pad-to-hole margin too small
- ▶ Hole-to-hole spacing: drill bits would collide
- ▶ Silkscreen on pad: silk covering solderable area
- ▶ Courtyard overlap: two components physically interfere
- ▶ Unconnected nets: ratsnest lines not routed

Net Classes

- ▶ Group nets by purpose: 'Power', 'Signal', 'HighSpeed', 'Differential'
- ▶ Each net class has its own: clearance, trace width, via size
- ▶ Power class: wider traces (e.g., 20 mil), larger via (0.6 mm drill)
- ▶ High-speed class: tighter trace width, controlled impedance
- ▶ Differential class: pair-coupling settings, length match tolerance
- ▶ Assign nets in schematic: Tools → Edit Symbol Properties → Net Class
- ▶ Or via design rules: 'add net SDA to Class HighSpeedI2C'
- ▶ Multiple classes per project — typical 3-5 classes

KiCad 8/9 Custom DRC Rules

.kicad_dru file in project folder – KiCad's custom rule language

(version 1)

Min clearance for high-speed traces

```
(rule "HighSpeed clearance"  
  (constraint clearance (min 10 mil))  
  (condition "A.NetClass == 'HighSpeed'"))
```

DDR4 byte-lane match within 30 ps

```
(rule "DDR4 length match"  
  (constraint length (min 90 mm) (max 95 mm))  
  (condition "A.NetClass == 'DDR_DQ0'"))
```

Different rule for analog section

```
(rule "Analog separation"  
  (constraint clearance (min 20 mil))  
  (condition "A.NetClass == 'Analog' && B.NetClass == 'Digital'"))
```

Constraint Types



Constraint	Example use
clearance	Min spacing between copper
track_width	Min trace width
via_diameter	Min via diameter
via_drill	Min via drill hole
length	Trace length match
skew	Differential pair skew
edge_clearance	Min copper to board edge
physical_clearance	Min spacing between footprints
hole_clearance	Min copper to hole edge
silk_clearance	Min silk to pad clearance
text_thickness	Min silk text line width

Conditional Rules

- ▶ Conditions use S-expression syntax matching KiCad's internal model
- ▶ Net-class-based: `'A.NetClass == "Power"'`
- ▶ Layer-based: `'A.Layer == "F.Cu"'`
- ▶ Component-type: `'A.Type == "SMD"'`
- ▶ Footprint-based: `'A.Parent.Reference == "U1"'`
- ▶ Combine with `&&` (AND) or `||` (OR): `'A.NetClass == "Power" && A.Layer == "F.Cu"'`
- ▶ Use `!A` for negation
- ▶ Test rules: run DRC — failures appear with rule names

Layer-Specific Rules

- ▶ Different rules for different layers (outer vs inner)
- ▶ Example: 6 mil min trace on outer, 4 mil on inner
- ▶ Outer layer: tighter clearance (more parts, more dense)
- ▶ Inner layers: looser clearance (mostly planes)
- ▶ BGA escape layers (inner): require microvia + tight rules
- ▶ Power planes: special rules (e.g., width = full pour, ground stitching density)
- ▶ Custom rule: `A.Layer` matches the layer name
- ▶ Stackup-specific: `'A.Layer == "L3"'` for layer 3 specific rules

Differential Pair Rules

- ▶ Diff pair coupling: gap should equal trace width (typical)
- ▶ Skew constraint: max allowed length difference within pair
- ▶ Layer constraint: both traces same layer (always)
- ▶ Symmetric routing rules: harder to encode in DRC, mostly visual + check
- ▶ Example for USB diff pair:
 - ▶ (constraint diff_pair_gap (min 0.13 mm))
 - ▶ (constraint diff_pair_uncoupled_length (max 5 mm))
- ▶ Net class 'USB_HS' applied to both D+ and D- nets

Importing Fab Rules

- ▶ Major fabs publish their capability matrix
- ▶ Match KiCad design rules to YOUR fab's spec
- ▶ JLCPCB: min trace 5 mil, min spacing 5 mil, min drill 0.3 mm
- ▶ PCBPower: similar, slightly different min spec
- ▶ Don't push to fab's minimum — leave margin (8 mil trace if fab supports 6 mil)
- ▶ DFM is two-sided: your rules > fab's min + you have margin
- ▶ If fab can't hold your spec, board gets rejected at QC

Common Rule Mistakes

- ▶ Rules too tight: hand-routing impossible, DRC errors everywhere
- ▶ Rules too loose: production-bad designs pass DRC
- ▶ No net class for high-speed → standard rules used (wrong for SI)
- ▶ Forgetting silk-on-pad rule → silk obscures pads at assembly
- ▶ No edge clearance rule → traces too close to board edge → manufacturing fails
- ▶ Missing courtyard rule → components physically interfere
- ▶ Custom rule syntax errors → KiCad logs warning but rule not applied
- ▶ Updating rules but not re-running DRC after change

Sharing Rules Across Projects

- ▶ Rules in `.kicad_dru` file at project root — text, Git-friendly
- ▶ Copy rules file between projects to share standards
- ▶ Or: link rules file from a shared corporate repo
- ▶ Document each rule with comment explaining WHY
- ▶ Version-tag rules: 'v1.0' through 'v2.0' as practices evolve
- ▶ On new project: start from corporate template rules → customize
- ▶ CI: validate rules file syntax + test on a baseline project

Key Takeaways

Key takeaways from this lecture

Codify design intent

Rules > hand-tracking errors.

Net classes by function

Power, signal, high-speed, diff.

Custom DRC powerful

S-expr rules for any condition.

Match fab capability

Margin above min spec.

Layer-specific rules

Outer vs inner often differs.

Share across team

Git-tracked .kicad_dru file.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ docs.kicad.org — Custom DRC rules section
- ▶ KiCad 8 release notes — custom DRC syntax
- ▶ JLCPCB / PCBWay capability matrices
- ▶ Phil's Lab YouTube — custom rules tutorials
- ▶ INTI-CMNB DRC examples
- ▶ Real-world rule sets (open-source PCB projects)

Thank you • Build something this week.